

Exclusion of the fixed-quadratic primitive-pencil quartic row

A frozen-case structural theorem for Keller maps in dimension three

Alec Kriebel
with substantial AI assistance

First public repository release (UTC): 2026-07-26T00:45:00Z

Review, priority, and verification notice. This manuscript is not peer reviewed. Exact computer algebra checks the identities and finite case ledgers encoded in the accompanying repository; it is evidence, not peer review. Substantial AI assistance was used in the mathematical exploration, proof organization, software, hostile internal audits, and exposition. No claim of worldwide priority is made. The result excludes one of fourteen frozen quartic leading-form rows; it does not prove the full degree-four case.

Abstract

Let $F : \mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ be a polynomial map of total degree four with nonzero constant Jacobian. We prove that F is an automorphism when its quartic homogeneous part has a fixed quadratic component and a primitive quadratic line pencil. In the program's canonical notation, the excluded row is

$$(\text{rank } JH_4, e, a, b, \delta, \nu) = (2, 2, 2, 1, 1, 1).$$

The proof first gives a coefficient-independent normalization $H_4 = (hp, hq, 0)$ and uses valuations of homogeneous first integrals to reduce to a unique-double-line frontier. Keeping separate the fixed quadratic h , the double member s , and the cubic companion produces a genuine $3 + \mathbb{P}^1 + 3$ orbit space, rather than a finite companion list. Exact weighted-Jacobian identities exclude the whole projective family, its six endpoints, and two discrete open orbits. A clean-room dependency-free reconstruction verifies the frozen coefficient routing and the two discrete calculations. The universal certified total-degree floor remains four.

1 Invariant statement

Translate the source and target so that

$$F = LX + H_2 + H_3 + H_4, \quad L \in \text{GL}_3(\mathbb{C}), \quad (1)$$

where H_i is homogeneous of degree i . Suppose that $\text{rank } JH_4 = 2$. Extract

$$h = \gcd(H_{4,1}, H_{4,2}, H_{4,3}), \quad e = \deg h, \quad G = H_4/h.$$

Let E_G be the relative algebraic closure of $\mathbb{C}(G_i/G_j)$ in $\mathbb{C}(\mathbb{P}^2)$. It is rational, so $E_G = \mathbb{C}(p/q)$ for a coprime homogeneous pair of least common degree a . If the induced binary target triple has degree b , the reduced image has degree δ , and the map to its normalization has degree ν , then

$$e + ab = 4, \quad b = \delta\nu.$$

Theorem 1 (Fixed-quadratic primitive-pencil exclusion). *Let F be as in (1), over $\mathbb{C}$, of exact total degree four. Suppose*

$$(\text{rank } JH_4, e, a, b, \delta, \nu) = (2, 2, 2, 1, 1, 1).$$

If F has nonzero constant Jacobian determinant, then F is a polynomial automorphism.

Corollary 2. *No degree-four Keller counterexample belongs to frozen row Q2-E2-A2-B1-D1-N1.*

The fourteen-row frozen denominator currently has

$$4/14 \text{ certified excluded}, \quad 4/14 \text{ provisional}, \quad 6/14 \text{ open}.$$

This bookkeeping is not a statement about the other rows.

2 Normalization and weighted identities

Proposition 3. *After an invertible target change,*

$$H_4 = (hp, hq, 0), \tag{2}$$

where h, p, q are quadratic, $\gcd(p, q) = 1$, and $\mathbb{C}(p/q)$ is relatively algebraically closed in $\mathbb{C}(\mathbb{P}^2)$. Every frozen coefficient chart is covered without dividing by its first nonzero quartic coefficient.

Proof. The canonical factorization is $H_4 = hA(p, q)$, where $A(u, v) = \mathbf{a}u + \mathbf{b}v$ and $\mathbf{a}, \mathbf{b} \in \mathbb{C}^3$ are independent. Complete them to a target basis. This gives (2) and transports all lower jets bijectively.

For a division-free coverage statement, write the coefficients of the primitive quadratic triple as a 3×6 matrix B . Its rank is two. Order its $45 = 3\binom{6}{2}$ minors and choose the first nonzero one. The corresponding two target rows span the third, and the cleared row relation has determinant equal, up to sign, to the selected minor. Thus it gives (2). Among the frozen first-nonzero coefficient guards, C00–C29 reach this minor router. The guards C30–C44 make the first two target components zero, so $\text{rank } JH_4 \leq 1$; they are empty in the present row. $\square$

Put

$$\mathcal{J}(w) = L + wJH_2 + w^2JH_3 + w^3JH_4, \quad E_j = [w^j] \det \mathcal{J}(w).$$

The Keller condition gives

$$E_9 = \cdots = E_1 = 0, \quad \det L \neq 0. \tag{3}$$

Writing $P = hp$, $Q = hq$, and $R = (H_3)_3$, the zero third row of JH_4 gives

$$E_8 = \text{Jac}(P, Q, R) = 0. \tag{4}$$

3 The top-identity frontier

Lemma 4 (First-integral descent). *If $0 \neq T$ is homogeneous of degree d and $\text{Jac}(P, Q, T) = 0$, then*

$$\frac{T^4}{P^d} = \mathcal{R}(q/p)$$

for some $\mathcal{R} \in \mathbb{C}(t)$.

Proof. The Jacobian identity makes T algebraic over $\mathbb{C}(P, Q)$. The displayed quotient has degree zero, hence lies in $\mathbb{C}(\mathbb{P}^2)$. Separating the scaling coordinate shows that it is algebraic over $\mathbb{C}(q/p)$. Relative algebraic closure of the minimal pencil field then places it in $\mathbb{C}(q/p)$. $\square$

A prime factor f of h is called horizontal if it is not contained in a pencil fibre. Taking v_f in Lemma 4, with $d = 3$, gives

$$4v_f(R) = 3v_f(h).$$

Since $v_f(h)$ is one or two, this is impossible for nonzero R . Thus a horizontal component forces $R = 0$.

For the all-vertical case, comparing two reduced components of one pencil fibre removes the unknown order of $\mathcal{R}$. This yields the following exact reduction.

Proposition 5 (All-vertical reduction). *If every component of h is vertical and (4) has a nonzero cubic solution, then, after source and pencil changes,*

$$H_4 = (h^2, hs, 0), \quad s = \ell^2, \quad R = \ell r, \quad [r] \in \mathbb{P}\langle h, s \rangle, \quad (5)$$

where s is the pencil's unique double-line member. The underlying pencil is one of

$$\langle x^2, yz \rangle, \quad \langle x^2, y^2 + xz \rangle.$$

Proof. If $h = \ell^2$ and a vertical member is ℓm with $m \not\sim \ell$, comparison of the two components of that fibre gives $4N = 6$. If $h = \ell_1 \ell_2$ and the vertical members are distinct, the same argument gives $4N = 3$ unless both members are squares; the latter would make p/q a quadratic function of a linear pencil and contradict minimality.

The remaining shape has one pencil member equal to h . Then

$$\text{Jac}(h^2, hq, R) = 2h^2 \text{Jac}(h, q, R).$$

First-integral descent gives $R^2/h^3 = S(q/h)$. If every conic fibre were reduced, all finite orders of S would be even and the order at the h -fibre odd, contradicting degree zero of a principal divisor on $\mathbb{P}^1$. Hence the pencil contains a double line $s = \ell^2$. Two distinct double members would again violate minimality, so s is unique. Direct calculation gives the complete cubic kernel $\ell\langle h, s \rangle$.

Put $s = x^2$. The restriction of another member to $x = 0$ has rank two or one. Completing cross terms gives respectively $\langle x^2, yz \rangle$ and $\langle x^2, y^2 + xz \rangle$. This proves the claim. $\square$

Lemma 6 (Quadratic-component exit). *If a degree-at-most-four Keller map of $\mathbb{A}_{\mathbb{C}}^3$ has a nonzero target-linear component of degree at most two, it is an automorphism.*

Proof. A quadratic polynomial with nowhere-vanishing gradient can be straightened to a coordinate by a polynomial automorphism of degree at most two. The transformed map is (F_1, F_2, z) . On every z -fibre the first two components form a plane etale map of degree at most eight. Moh's unconditional bounded plane theorem, in the degree-at-most-twelve form quoted by Vistoli, makes every fibre map an automorphism [1, 2]. Hence the threefold map is injective and therefore an automorphism. $\square$

Consequently every branch with $R = 0$ exits the counterexample locus by Lemma 6. It is important that this is an automorphism conclusion, not a contradiction to the existence of a constant Jacobian.

4 The marked companion quotient

The frontier (5) has three pieces of intrinsic data: the pencil, its double member $[s]$, the fixed gcd $[h]$, and the companion $[r]$. The loci $h = s$ and $h \neq s$ must be treated separately.

When $h = s$, the two canonical pencils above have exactly two nonzero companion orbits:

$$R = x^3, \quad R = xq.$$

Exact weighted-determinant eliminations exclude both orbits for both pencils. These four calculations are the marked-equal ancillary packages.

For $h \neq s$, the marked pair and companion have exactly the following thirteen stable strata.

marked pair (s, h)	companion suffix	representative or guard
(x^2, yz)	C0	$R = 0$
	CH	$r = h$
	CS	$r = s$
	CO	$r \notin \{h, s\}$
$(x^2, x^2 + yz)$	C0	$R = 0$
	CH	$r = h$
	CT	$r = yz$
	CS	$r = s$
	CTAU	$r = h + ks, \ k \neq 0, -1$
$(x^2, y^2 + xz)$	C0	$R = 0$
	CH	$r = h$
	CS	$r = s$
	CO	$r \notin \{h, s\}$

The middle stabilizer fixes its pencil line pointwise, so CTAU is a genuine one-parameter orbit family. The projective values $0, -1, \infty$ are, respectively, CH, CT, and CS; none may be removed by a generic calculation. Two independent stabilizer reconstructions certify the count

$$4 + 5 + 4 = 13, \quad \text{nonzero orbit-space shorthand } 3 + \mathbb{P}^1 + 3.$$

5 Uniform exclusion of the projective family

On the finite chart put

$$h = x^2 + yz, \quad s = x^2, \quad r = h + ks, \quad k \neq 0.$$

The complete E_7 normal form, modulo two target shears and three source translations, is

$$H_3 = (Ax^3, Bx^3, x(h + ks)), \quad (H_2)_3 = Tx^2.$$

Two E_7 pivots cover every finite k because

$$\frac{1}{2}(9k^2 + 6k - 1) - \frac{3}{2}(k + 1)(3k - 1) = 1.$$

Write the first two quadratic components in monomial order $(x^2, xy, xz, y^2, yz, z^2)$ with coefficients a_i, b_i , and put $L = (\ell_0, \dots, \ell_8)$ by rows. The E_6 chains give

$$\begin{aligned} b_1 &= 2\ell_7, & a_1 &= -12k\ell_7, & -36k^2\ell_7 &= 0, \\ b_2 &= 2\ell_8, & a_2 &= -12k\ell_8, & 36k^2\ell_8 &= 0, \end{aligned}$$

together with $a_3 = a_5 = b_3 = b_5 = 0$. Since $k \neq 0$,

$$a_1 = a_2 = a_3 = a_5 = b_1 = b_2 = b_3 = b_5 = \ell_7 = \ell_8 = 0.$$

Six literal E_5 coefficients then include

$$-2\ell_4, \quad 2\ell_5, \quad -\ell_1 - (6k+4)\ell_4, \quad \ell_2 + (6k+4)\ell_5,$$

so $\ell_1 = \ell_2 = \ell_4 = \ell_5 = 0$. The last two columns of L vanish, contrary to (3). This uniformly excludes CTAU and the finite CT boundary $k = -1$.

6 Six endpoints and two discrete orbits

The six CH/CS endpoint normal forms have four E_6 compatibility ideals:

type	I_6
RT/H	$(AC, AD, AE, AF, CE, DF, E^2, F^2)$
RT/S	(C^2, D^2)
RO/H	$(AC, AD, AE, AF, CF + DE, EF, 2DF - E^2, F^2)$
RO/S	(CD, D^2) .

Global E_5 syzygies and fresh solves on every pivot divisor force a singular L in five endpoint slices. The sixth, RO/H, has a genuine invertible survivor through E_5 on

$$A = C = D = E = F = 0, \quad T \neq 0.$$

After the complete E_6/E_5 solve, two E_4 coefficients are

$$[xyz^2]E_4 = -8\ell_8^2, \quad [x^2yz]E_4 = -4(2(b_0 - \ell_6)\ell_8 - \ell_7^2).$$

They give $\ell_8 = \ell_7 = 0$, while $\det L$ is divisible by ℓ_7 . Thus all six endpoints close.

It remains to record the two outer CO calculations. For

$$(h, r) = (yz, x^2 + yz)$$

the raw E_7 matrix has rank 18, with five legal gauge directions and complete normal complement

$$(x^3, 0, 0), \quad (0, x^3, 0), \quad (0, 0, x^2).$$

The E_6 equations force

$$a_1 = a_2 = a_3 = a_5 = b_1 = b_2 = b_3 = b_5 = \ell_7 = \ell_8 = 0,$$

and E_5 forces $\ell_1 = \ell_2 = \ell_4 = \ell_5 = 0$.

For

$$(h, r) = (y^2 + xz, x^2 + y^2 + xz)$$

the complete normal complement is

$$(x^3, 0, 0), \quad (0, x^3, 0), \quad (2z(y^2 + xz), x^2z, xz).$$

If C is the last normal parameter, the E_6 equations give

$$a_1 = a_4 = b_1 = b_4 = b_5 = \ell_7 = \ell_8 = 0, \quad a_2 = a_3, \quad a_5 = C^2, \quad b_2 = b_3.$$

Then E_5 gives

$$\ell_1 = \ell_4 = 0, \quad \ell_2 = Ca_3, \quad \ell_5 = Cb_3.$$

The second column of L is zero. This calculation includes $C = 0$ and uses no localization.

7 Assembly of the proof

Proof of Theorem 1. Proposition 3 covers every frozen coefficient guard. If h has a horizontal component, the valuation congruence forces $R = 0$, and Lemma 6 gives an automorphism. If every component is vertical, Proposition 5 either gives the same exit or reaches (5).

On the frontier, $R = 0$ again gives an automorphism. For $R \neq 0$, the marked-equal packages or exactly one of the thirteen marked-distinct strata applies. The four marked-equal nonzero orbits, the uniform projective family, all six endpoints, and both discrete CO orbits force $\det L = 0$. This contradicts the constant term of the Keller determinant. No Keller counterexample remains in the row. $\square$

8 Verification and audit

The aggregate command

```
./verify_full_row_strict.sh
```

runs exact SymPy, independent PARI/GP, and dependency-free checks for the top reduction, all lower packages, and the post-freeze bridge. Its final marker is

```
Q2_E2_A2_B1_D1_N1_FULL_ROW_STRICT_PASS_4D95A1
```

The clean-room bridge auditor did not read the candidate bridge, the root exploration scripts, or the primary CO and endpoint verifiers. Its standard-library sparse-polynomial engine independently reconstructs:

30 rank-two-possible frozen pivots, 15 empty pivots, 45 intrinsic minor charts,

the $4 + 5 + 4$ marked ledger, both raw CO quotients, their lower exits, and the uniform finite chart. Its marker is

```
AUDIT_BRIDGE_Q2_E2_STRICT_PASS_D9347B
```

The audit found and corrected two draft errors: C0 is an automorphism exit rather than a blanket determinant contradiction, and C30–C44 are empty rather than entering the minor router. Exact checks verify the encoded algebra; they are not peer review.

References

- [1] T.-T. Moh, *On the Jacobian conjecture and the configurations of roots*, J. Reine Angew. Math. **340** (1983), 140–212.
- [2] Angelo Vistoli, *The Jacobian conjecture in dimension 3 and degree 3*, J. Pure Appl. Algebra **142** (1999), 79–89.